

Tatevik Mkrtchyan

PhD Student in Astrophysics, Universidad Diego Portales, Santiago, Chile
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RESEARCH INTERESTS

Active Galactic Nuclei, High-redshift quasars, Galaxy Evolution and Formation, Supermassive Black Holes, Cosmology, Surveys

EDUCATION

Universidad Diego Portales, Santiago, Chile August 2022 — Present
PhD Student in Astrophysics
Thesis Title: Characterizing Active Galactic Nuclei across Cosmic Time

Yerevan State University, Yerevan, Armenia September 2019 — May 2021
Master of Science in Astrophysics
Thesis Title: The Study of High Energy Gamma-Ray Emission from Galaxy Clusters

Yerevan State University, Yerevan, Armenia September 2014 — June 2019
Bachelor of Science in Physics
Thesis Title: High Energy Gamma Ray Radiation Processes of Blazars

ACADEMIC EXPERIENCE

Universidad Diego Portales Santiago, Chile
PhD Student in Astrophysics August 2022 — Present

- Characterizing high-redshift quasars using SED fitting for 4MOST.
- Identifying Lyman alpha halos and studying the 3D maps of them
- Developing photometric selection methods of AGN with LSST.

Byurakan Astrophysical Observatory Byurakan, Armenia
Senior Lab Assistant April 2021 — 2022

- Worked on Galaxy Evolution models using Principal Component Analysis in the AGN group.

International Center for Relativistic Astrophysics Yerevan, Armenia
Junior Researcher Intern 2016 — 2018

- Made models of high-energy gamma-ray radiation processes of blazars using Wolfram Mathematica

PUBLICATIONS

In Progress and Submitted

- 4MOST ChANGES: Catalog of high-redshift quasar candidates selected with SED fitting (Mkrtchyan et al. 2025, submitted to AA)
- Inclusive Teaching Methods Handbook: Published by The Republican Pedagogical-Psychological Center of RA

For a complete list of publications, visit: [ADS Publication Profile](#)

OBSERVATIONS

11225VZ002 in the 3.6-metre telescope La Serena, Chile
La Silla Observatory, 4 nights January 2024

6M0B5WI1 Observing run in the 6.5-meter Baade Magellan Telescope La Serena, Chile
Las Campanas Observatory, 2 nights August 2023

0914WWG7 Observing run in the 6.5-meter Baade Magellan Telescope La Serena, Chile
Las Campanas Observatory, 1 night May 2023

XVJ9B6M2 Observing run in the 6.5-meter Baade Magellan Telescope
Las Campanas Observatory, 2 nights

La Serena, Chile
 April 2023

NON-ACADEMIC EXPERIENCE

National Center of Educational Technologies
Education Expert/Lead Physics Teacher

Yerevan, Armenia
 2020 — 2022

- Mentored teachers in implementing remote teaching methods, developed quality assurance tools for educational programs
- Made 100+ videos based on a physics school program available on YouTube.

State Pedagogical-Psychological Center
Teacher Trainer/Mentor

Yerevan, Armenia
 2019 — 2022

- Mentored educators in effective classroom management techniques and wrote a handbook on STEM teaching methods.

Shirakatsy Lyceum International Scientific-Educational Complex
Physics/Astrophysics Teacher

Armenia
 2018 — 2019

- Led "Young Astrophysicist" student club and mentored students in scientific research projects

TUMO Center for Creative Technologies
Robotics Workshop Leader

Armenia
 2017 — 2018

- Taught robotics to children aged 12-18 and developed STEM curriculum.

CONFERENCES AND PRESENTATIONS

SOCHIAS XX Annual Meeting [Oral Presentation]
 SOCHIAS/ Universidad Austral de Chile

Oct 2025

SOCHIAS XX Annual Meeting [Poster]
 SOCHIAS/ Universidad Austral de Chile

Oct 2025

Extreme galaxies in their extreme environments at extremely early epochs [Poster]
 The National Institute for Astrophysics

May 2024

The Santiago AGN workshop: a holistic view of active galaxies [Oral Presentation]
 European Southern Observatory

November 2023

18th Heidelberg Summer School [Participant]
 Max Planck Institute for Astronomy, University of Heidelberg

September 2023

New Era of AGN Science with the Vera C. Rubin LSST [Oral Presentation]
 LSST AGN Science Collaboration, National Radio Astronomy Observatory

July 2023

Science Leadership Conference [Participant]
 The Optical Society (Optica)

September 2020

VHE Phenomena around Supermassive Black Holes Workshop [LOC]
 Russian and Armenian University, University of Heidelberg, Volkswagen Foundation

September 2019

Sixth Byurakan Summer School [Participant]
 Byurakan Astrophysical Observatory

September 2018

Wolfram Summer School [Participant]
 Wolfram Research Company

August 2017

IONS International Conference [LOC]
 OSA YSU and NAS Chapter

June 2017

SKILLS

- **Languages:** Armenian (Native), Russian (Fluent), English (Fluent), Spanish (Intermediate)
- **Technical:** Python (matplotlib, pandas, astropy), Wolfram Mathematica, LePhare, EsoReX, LaTeX
- **Soft Skills:** Scientific Communication, Teaching, Research Leadership, Project Management

AWARDS

Beca Adelina Gutiérrez Sociedad Chilena de Astronomía	Chile 2024
Finalist of Falling Walls Science Engage International science communication competition	Global 2020
Scholarship after Viktor Hambardzumyan Prestigious astronomy scholarship	Armenia 2020
Scholarship after Gevorg and Gohar Vardanian Academic excellence scholarship	Armenia 2014
Annual Presidential Award of the RA in IT Sphere Recognition for contributions to information technology	Armenia 2014

REFERENCES

Dr. Chiara Mazzucchelli

Supervisor, Department of Physics, Universidad Diego Portales, Santiago, Chile

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Additional references available upon request